Youchul Jeong

E-mail: youchul1821@gmail.com / **Phone**: (+82)10-4912-1821
LinkedIn: https://www.linkedin.com/in/youchul-jeong-9b1a3a194/
GitHub: https://github.com/youchulJ

Education

Pusan National University in Department of Earth Environmental System — Busan, South Korea
Master of Science in Oceanography — ***Feb 2024***

- MSc Thesis "A Study on Monitoring Floating Marine Macro Litter using a Multi-spectral Sensor and Classification based on Deep Learning." (Supervisor: Prof. Young-Heon Jo)
- Oral presentation award, Ocean & Atmosphere Science Interdisciplinary Studies Conference
- BrainKorea21 Scholarship (Earth Environmental Systems Education Research Group) — $5100
- GPA (4.25/4.5)

Pusan National University in Department of Oceanography — Busan, South Korea
Bachelor of Science in Oceanography — ***Feb 2022***

- BSc graduation paper "A Study on Image-Based Marine Litter Detection Techniques." (Supervisor: Prof. Young-Heon Jo)
- Center of Remote Sensing Lab — Jan 2021 - Present
- Academic Excellence Scholarship from the Department of Oceanography — $1700
- GPA (3.24/4.5)

Kunsan National University in Department of Marine Construction Engineering — Gusan, South Korea

- Academic Excellence Scholarship from the Department of Marine Construction Engineering — $5000
- Coursework before transferring to Pusan National University

Research Experience

Drone Analysis of Beach Safety and Beachgoer Behavior — ***Nov 2023, Feb - Aug 2024***

- Orchestrated drone flights and implemented the YOLOv8 algorithm to observe beach safety.
- Conducted surveys of beachgoer population to understand motivations and improve beach conditions.

Classification and Monitoring of Floating Marine Macro Litter (MSc Thesis) — ***Jun 2022 - Dec 2023***

- Utilized drones with multispectral sensors to create a custom FMML dataset.
- Implemented deep learning models for efficient litter classification and continuous monitoring.
- Incorporated drone atmospheric correction using DROACOR in collaboration with Dr. Daniel Schläpfer (ReSe Application LLC).
- Worked with advisors to promote conceptual framework and fieldwork at international conferences.

Floating Marine Debris and Physical Ocean Phenomena in S-ORS — ***April 2021 - Dec 2023***

- Applied drone technology and Applied Drone Automatic Flight Advisory in open ocean conditions.
- Developed a CNN model for detecting marine debris for Socheongcho Ocean Research Station.

Drone Survey of Spotted seal Population on Baekryeong Island *May 2022 - Nov 2023*

- Conducted drone surveys to assess spotted seal population on Baekryeong Island and gather high-resolution imagery, contributing crucial data to understanding spotted seal dynamics.

Floating Marine Litter, Korea Institute of Ocean Science & Technology *May 2023 - Oct 2023*

- Worked with Dr. Won Joon Shim (KIOST) to conduct on-site surveys of microplastics and floating marine litter.
- Compared and analyzed results of drone and multispectral sensor surveys.

Coastal Topography and Seabed Survey *Jul 2023 - Aug 2023*

- Conducted RTK-GPS and drone surveys to analyze coastal sand loss and seabed depth change.

UAV Team Leader, Korean Air Force *May 2016 - May 2018*

- Assembled military training UAVs, managed GPS element procurement, and led team activities.

Teaching Experience

Research Assistant *Sep 2022 - Dec 2022 / Sep 2023 - Feb 2024*

- Developed innovative methodologies including drones, AI, and multi-spectral sensors to contribute to published articles on remote sensing.

Interpretation of marine data [Course No. MS3200990] *1st semester 2023*

- Led fieldwork for 40 undergraduate students to acquire Sargassum multi-spectral images.
- Taught data processing using MATLAB as part of the project curriculum.

Coastal Practices Survey [Course No. MS3200295] *1st semester 2022, 2023*

- Delivered theoretical lectures on YOLOv5, covering installation process and algorithms.

Publications and Presentations

Youchul Jeong, Jisun Shin, Jong-Seok Lee, Ji-Yeon Baek, Daniel Schläpfer, Sin-Young Kim, Jin-Yong Jeong, Young-Heon Jo., "A Study on Monitoring Floating Marine Macro-Litter using a Multi-Spectral Sensor and Classification based on Deep Learning.", Remote Sensing, 2024.

Sin-Young Kim, Jong-Seok Lee, **Youchul Jeong**, Young-Heon Jo., "Decomposition of Submesoscale Ocean Wave and Current Derived from UAV-Based Observation.", Remote Sensing, 2024.

Youchul Jeong et al., "A Study on Monitoring Floating Marine Macro Litter using a Multi-spectral Sensor and Classification based on Deep Learning." Poster presentation at 11th Asian Workshop on Ocean Color, Nagoya, Japan, 2023.

Youchul Jeong et al., "A Study on Monitoring Floating Marine Macro Litter using Multi-spectral Sensor and Classification based on Deep Learning." Oral presentation at Korea Ocean Research Conference, Seoul, South Korea, 2023.

Youchul Jeong et al., "A Study on Monitoring Floating Marine Macro Litter using a Multi-spectral Sensor and Classification based on Deep Learning." Oral presentation at Ocean & Atmosphere Science Interdisciplinary Studies Conference, Busan, South Korea, 2023.

Youchul Jeong et al., "A Study on Classification and Monitoring of Floating Marine Macro Litter using Multispectral Imagery in UAV and Deep Learning." Poster presentation at European Geoscience Union General Assembly, Vienna, Austria, 2023.

Youchul Jeong et al., "A Study on Classification and Monitoring of Floating Marine Macro Litter using Multispectral Imagery in UAV and Deep Learning." Oral presentation at 10th Asian Workshop on Ocean Color, Indonesia, 2022.

Volunteer Activities

Mentoring for Multicultural and North Korean refugee students ***Sep 2020 - Jan 2021***

- Taught Korean literature and science to a second-year middle school student from South Africa

Association Membership

- The Korean Society of Oceanography
- The Korean Society of Remote Sensing
- GeoAI Data Society
- Asian Workshop on Ocean Color
- European Geosciences Union General Assembly
- Ocean & Atmosphere Science Interdisciplinary Studies Conference

Relevant Academic Coursework

- Physical Oceanography
- Multiple Remote Sensing Environment
- Advanced Earth and Environmental System (Ⅰ), (Ⅱ)
- Biological Oceanography
- Python for Data Analysis
- Advanced Big Data Application for Earth System (Ⅱ)
- Chemical Oceanography
- Ocean Survey and Research Cruise
- Floating Biology and Experiments

Technical and Quantitative Skills

- MATLAB, Python, ENVI 5.2, and DROACOR (Drone Data Atmospheric Correction processor).
- 100+ UAV research flights with highest South Korean license (Drone weight 25 - 150kg).
- Bilingual in Korean and English.